AI-Driven Reservoir Performance Proxy

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Team Introductions

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Executive Summary

Business Problem: Traditional physics simulators predict oil reservoir output but are extremely slow (a single scenario can take minutes to hours).

- Over 1.1 million grid cells, each requiring fluid-flow calculations at every timestep

Project Goal: Build a Machine Learning Proxy that learns reservoir physics from simulation runs, then delivers new predictions in milliseconds. This would enable a real-time decision tool where users adjust sliders and instantly see a 20-year oil production forecast

Research Question: Does adding physics-aware constraints produce better, more realistic predictions than a purely data-driven model?

SPE10 Model & OPM Flow

- SPE10 (Model 2) is a well-known public benchmark from the SPE Comparative Solution Project, widely used in academic reservoir simulation
- Features a heterogeneous reservoir with defined grid properties, permeability variation, and well configurations
- OPM Flow serves as the physics engine, modeling how oil, gas, and water move through rock using complex fluid-flow equations
- The simulated reservoir timeline spans 8,000 days (~22 years) of production life
- Key outputs tracked: Field Oil Production Rate (FOPR), Cumulative Oil Production (FOPT), Field Pressure (FPR), and Bottom Hole Pressure (BHP)

Data Overview

- 200 unique simulation scenarios generated using Latin Hypercube Sampling (LHS) to ensure broad, efficient coverage of the input space
- Runs 1–100 (Operational Sensitivity): One variable changed at a time to isolate individual effects on production
- Runs 101–150 (Chaotic Group): All settings varied simultaneously to teach the model how inputs interact with each other
- Runs 151–200 (Geological Group): Rock properties (porosity and permeability) varied to capture subsurface uncertainty
- Inputs include: permeability/porosity fields, well-control variables (injection rates, BHP), time-step data, and prior reservoir states
- Outputs include: pressure distribution, water saturation, oil/water production rates, and BHP at wells

Random Forest

Role — fast baseline that sets the accuracy bar and ranks which inputs matter, before MLP / PINN / LSTM.

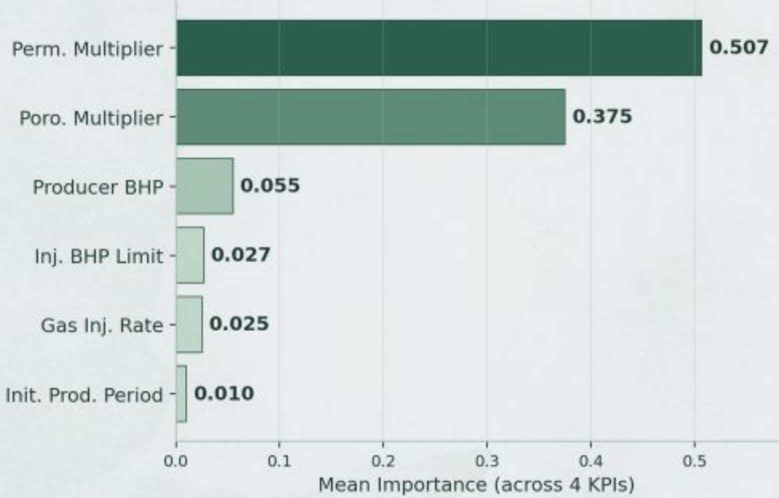
Setup — 6 inputs → 4 KPIs · 200 runs (160 train / 40 test) · 500 trees · 5-fold CV.

Baseline Accuracy

Target	Test R^2	MAE
Cumul. Oil — FOPT	0.988	499 STB
Final Oil Rate — FOPR	0.972	0.01 STB/d
Reservoir Press. — FPR	0.934	71 psi
Peak Oil Rate	0.982	12 STB/d

5-fold CV R^2 0.79–0.98 — metrics hold under resampling.

Feature Importance (mean across 4 KPIs)



Geology dominates: permeability + porosity drive ~88% of variance; operational controls register mainly through interactions.

Multilayer Perceptron

Input (6 features) → **Dense 128** (ReLU + BN + Drop(0.2)) → **Dense 64** (ReLU + BN + Drop(0.2)) → **Dense 32** (ReLU + BN + Drop(0.1)) → **Output (2 features)**

Inputs

- Bottom hole pressure (controllable)
- Gas Injection rate (controllable)
- Maximum pressure limit on the injector (controllable)
- Timing of the control phase switchover (controllable)
- Permeability multiplier (geological)
- Porosity multiplier (geological)

Outputs

- Total cumulative oil produced at Day 8,000
- Final average reservoir pressure at Day 8,000

Key Design Features

- Normalized with StandardScaler
- Adam optimizer (learning rate starting at 0.001)
- Dropout
- No output activation - final layer is linear
- Early stopping (patience=50)
- Batch size = 16
- ReduceLRONPlateau

Results:

final_fopt
Cumulative Oil Production
R^2 0.996
MAE
539.2 STB
Excellent

final_fpr
Final Reservoir Pressure
R^2 0.972
MAE
56.3 psi
Excellent

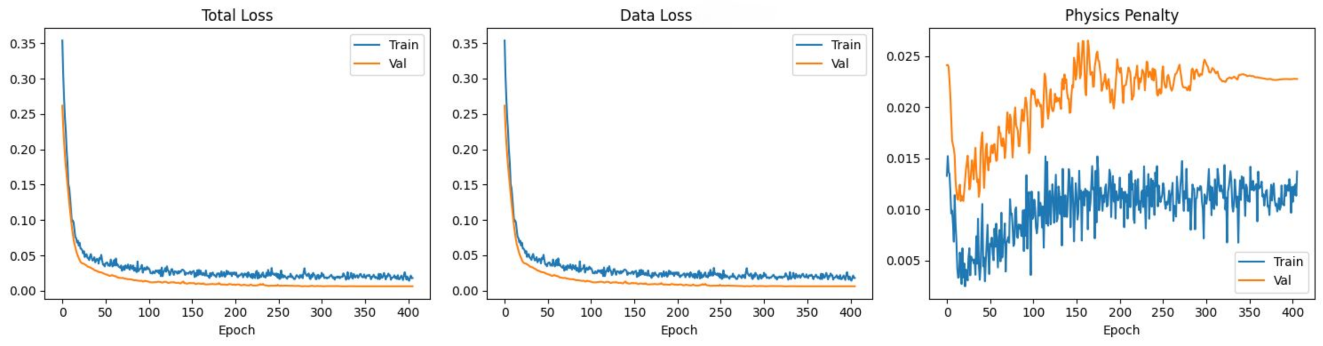
Physics-Informed Neural Network

Total Loss = Data Loss (Huber) + λ × Physics Penalty

Constraints:

- Darcy's Law
  - FPR > producer BHP
- Injection pressure ceiling
  - FPR ≤ injector BHP limit

Target	R^2	MAE
Final FOPT	0.996	526 STB
Final FPR	0.960	51.1 psi
Final FOPR	0.942	0.0167 STB/d
Peak FOPR	0.981	15.6 STB/d

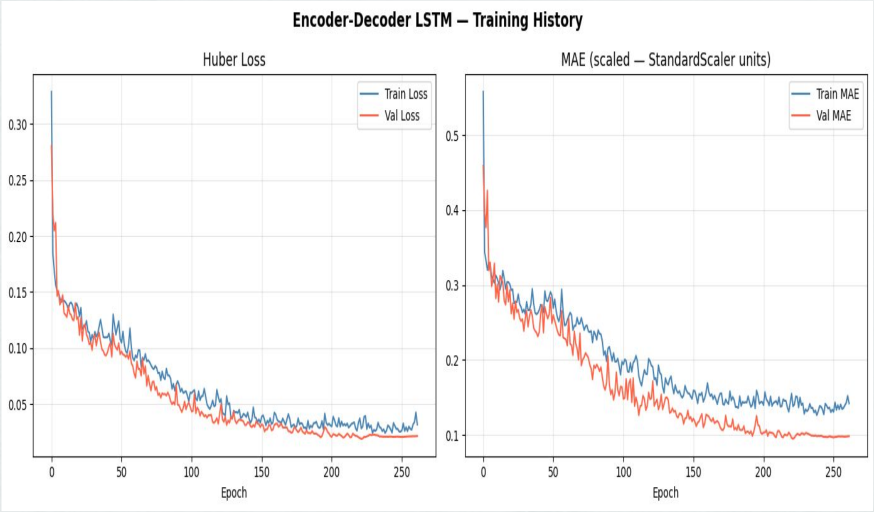


LSTM

The Limitation of Scalars:
- Predicting a single final number isn't enough. Reservoir engineers need to see the entire 22-year curve to calculate cash flows (NPV).

The Mathematical Hurdle:
- The physical simulator uses adaptive, irregular time steps (taking tiny steps when pressure changes rapidly).
- Standard neural networks cannot handle irregular sequence data efficiently.

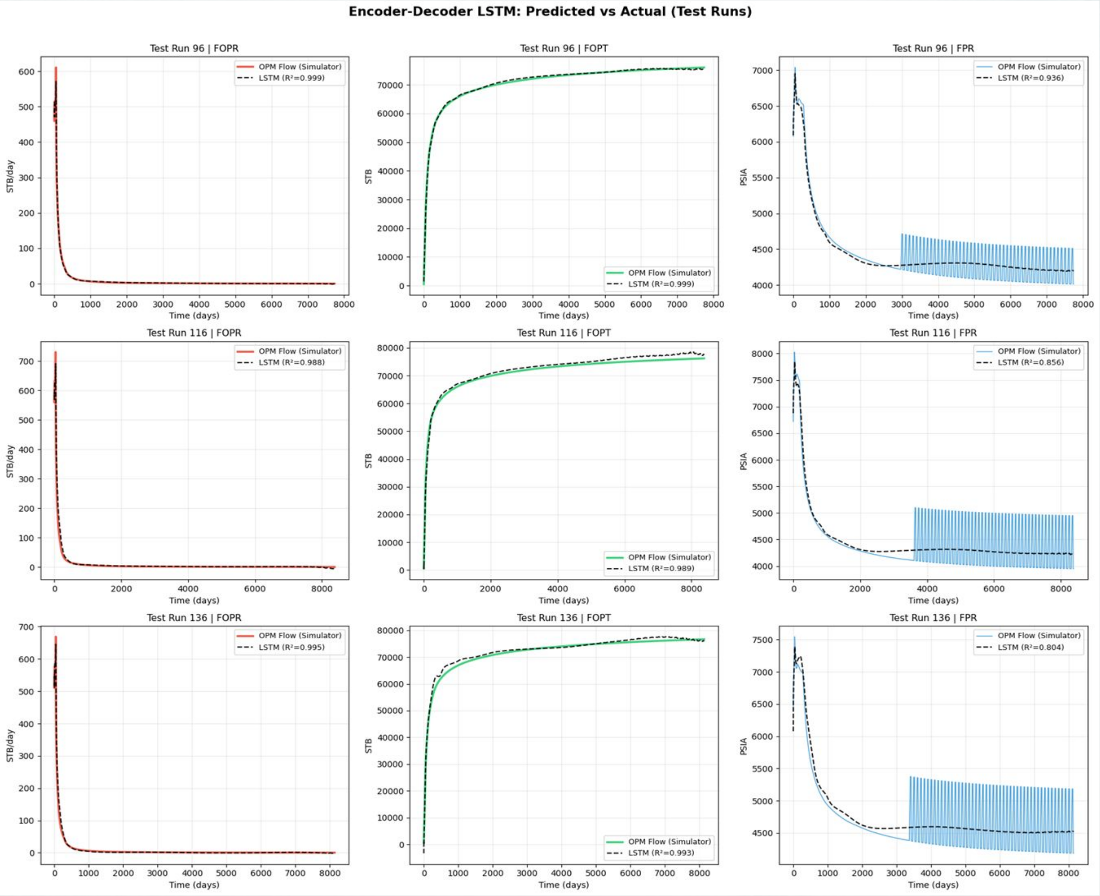
The Solution LSTM:
- 158,742 rows · 18 columns
- 200 runs × ~790 adaptive timesteps
- 8 physical outputs per timestep
- Padding + masking for variable lengths



LSTM

Why the LSTM is uniquely valuable:

- Predicts 8 physical variables across the full 22-year trajectory, not just final scalar values.
- FPR oscillations (R^2=0.842) reflect real cyclic gas injection physics captured by OPM Flow, model correctly learns the smooth depletion envelope without overfitting to operational noise.
- Adaptive average-grid inference allows prediction for any new scenario without rerunning the simulator



Conclusion & Insights

Best Models:

- **Physics Inform Neural Network (PINN):** PINN was selected over pure data-driven models because it uniquely enforces reservoir engineering laws during training guaranteeing 95% physical compliance across all test runs while matching MLP accuracy (fopt R^2=0.996), making it the only model safe for real engineering deployment.
- **Encoder-Decoder LSTM:** The Encoder-Decoder LSTM extends beyond scalar prediction to deliver full 22-year production curves for 8 physical variables simultaneously directly enabling the interactive reservoir management dashboard envisioned as future work.

Extreme Speed:

- What takes the physical simulator minutes, the AI evaluates in milliseconds.

Physical Reliability:

- By using PINNs and Loss Weighting, the predictions are fully compliant with thermodynamic laws.

Commercial Grade:

- This architecture bridges the gap between traditional petroleum engineering and cutting-edge deep learning.

Future Work

Increase size of the simulation dataset:

- Increasing the size of the dataset would help us improve model accuracy

Test out additional model types:

- Analyze performance on additional model types (Stacked models, Bayesian Neural Network, etc.)

Thank you